



clearmed

RELIABLE medical term simplifier for EVERYONE

The Problem & Target Audience

The Problem

Patients routinely leave clinical encounters holding **documents they cannot understand**. This gap creates high costs:

- **For Patients:** Increased anxiety and undermined ability to make informed decisions. (61% rate terminology as highly difficult to understand).
- **For Hospitals:** Preventable readmissions (a multi-billion dollar problem), repetitive follow-up calls, and wasted clinical staff time.

Target Audience

- **The Buyers (Hospitals & EMRs):** Departments responsible for discharge. EMR platforms (like OpenMRS) acting as distribution channels.
- **The Beneficiaries (Patients):** Individuals receiving care who need accessible instructions delivered directly to their phone.

Clinical Validation: Tested at Sheba & Rambam.

Physicians confirmed accuracy, time savings, and strong willingness to sign off on outputs.

Market, Competitors & Differentiation

Market & Competitors

- **Generative Translation (e.g., Vital):** Uses LLMs to write new text at runtime. High risk of medical hallucination.
- **Productivity Tools (e.g., Nabla):** Generates doctor's notes from audio. Built for clinician productivity, not patient literacy.

The ClearMed Advantage

- **No Generation at Runtime:** AI is used strictly as an offline selector. Explanations are verified source text.
- **EMR Embedded:** Operates inside existing tools (OpenMRS) rather than competing for standalone app adoption.
- **Deterministic Translation:** Supports 100+ languages on demand via Google Cloud Translation operating strictly on *already verified* text.

The Product: Two Workflows

Patient App (Self-Service)

A standalone web application built for the patient.

- Patients upload documents directly into their own secure, persistent library.
- Term simplification is triggered independently by the patient.
- Empowers the patient to review and annotate their files across multiple sessions.

OpenMRS Widget

A microfrontend embedded directly into the hospital's EMR.

- Designed to sit inside existing clinical workflows rather than replace them.
- Doctors review and approve simplified terms before delivery.
- Patients receive the verified document via a mobile-friendly QR code.

Core Features & Capabilities



EMR Embedded

Ships as a working microfrontend widget inside OpenMRS. It plugs directly into the patient chart, eliminating the need for clinicians to adopt standalone applications.



Deterministic Translation

Translates the final explanation document into 100+ languages via Google Cloud Translation operating strictly on already-verified text, ensuring safety for linguistic minorities.



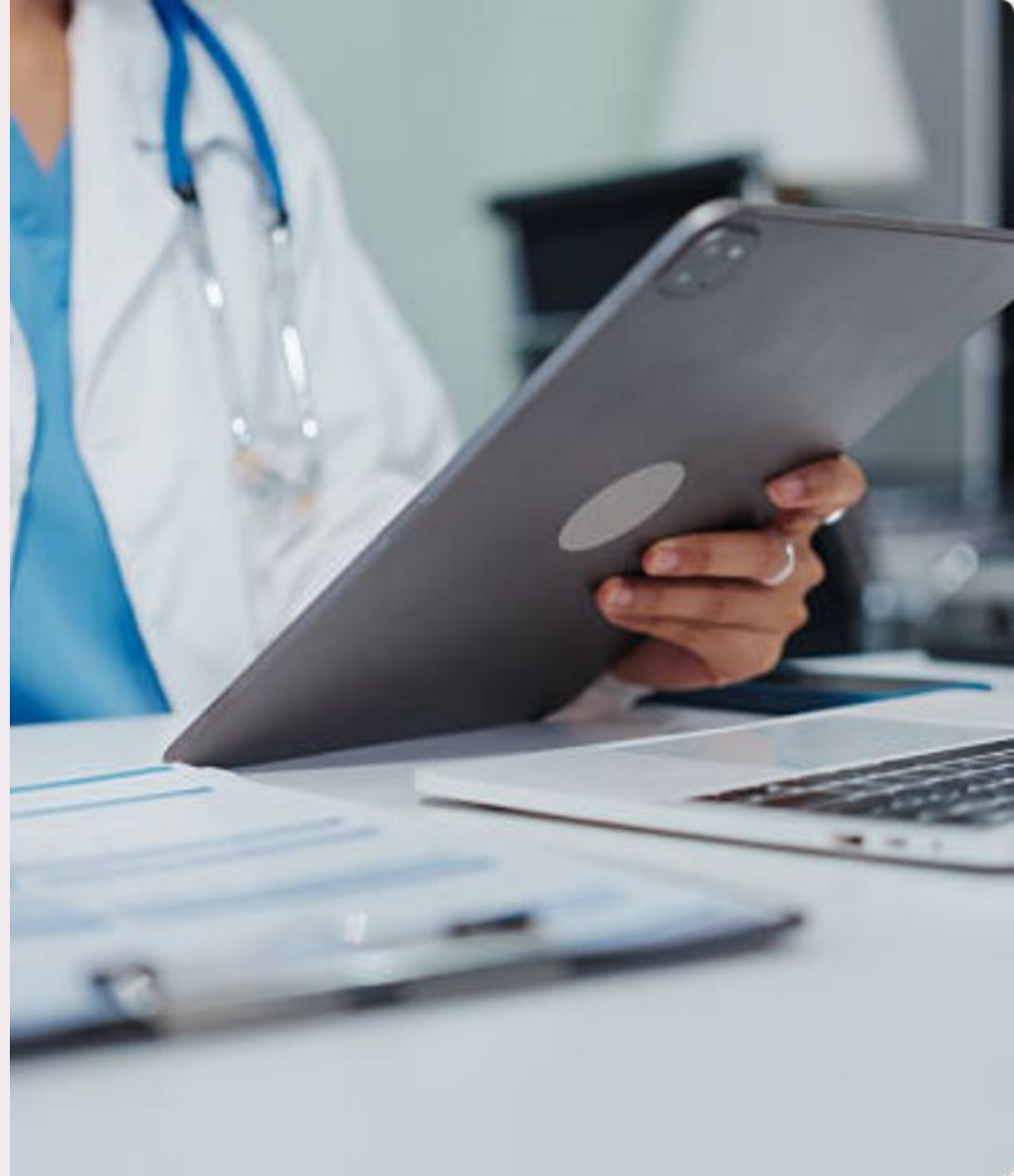
Hallucination-Free

Explanations are drawn sentence-for-sentence from MedlinePlus. AI acts strictly as an offline selector during database builds, never as a runtime writer.

OPENMRS integration

Doctor-in-the-Loop by Design

ClearMed is designed to sit inside existing hospital workflows. It detects medical terminology and attaches short, **source-verified explanations** (from MedlinePlus and Infomed). The doctor reviews and approves the result before it reaches the patient.



2. Term Detection

The system scans the extracted text against ClearMed's verified medical dictionaries, mapping concepts to MedlinePlus for English and Infomed for Hebrew.

4. Persistent medical record

The simplified explanations are saved directly to the original document in the patient's persistent library. Patients can reopen the document across different sessions, simplify different terms, or annotate the file.



1. Document Entry

Patients upload documents (PDF, JPG, or PNG) directly into their folders, which triggers an automatic OCR step for any scanned or photographed files..



3. simplification

Patients independently review the detected medical terms and select which ones they want explained. This workflow is entirely self-service and does not require doctors' review or approval.



Patient App: The Self-Service Pipeline

Database & Cloud Infrastructure



Managed PostgreSQL

User application data (accounts, folders, metadata) is stored in a dedicated Amazon RDS PostgreSQL database (clearmed_app). It is shielded from public access via strict EC2 security groups.



Abstracted Storage

Physical document files are managed through an abstract storage layer, seamlessly switching between local storage in dev and Amazon S3 for production



Secure S3 Integration

In production, files are securely stored in Amazon S3. Authentication relies entirely on an IAM instance role guaranteeing zero static AWS access keys in the deployment.

Fine-Tuning to shorten the long MedlinePlus paragraph

Platform: Together AI

Base Model: Qwen3.5-9B

Method: LoRA Fine-Tuning

Goal

MedlinePlus provides detailed medical information, often as long paragraphs. Our goal was to train a model to **select the single sentence that best explains a medical term in simple, patient-friendly language.**

How it works

For each medical term, the model receives:

Medical term + candidate sentences from its MedlinePlus text → Best sentence

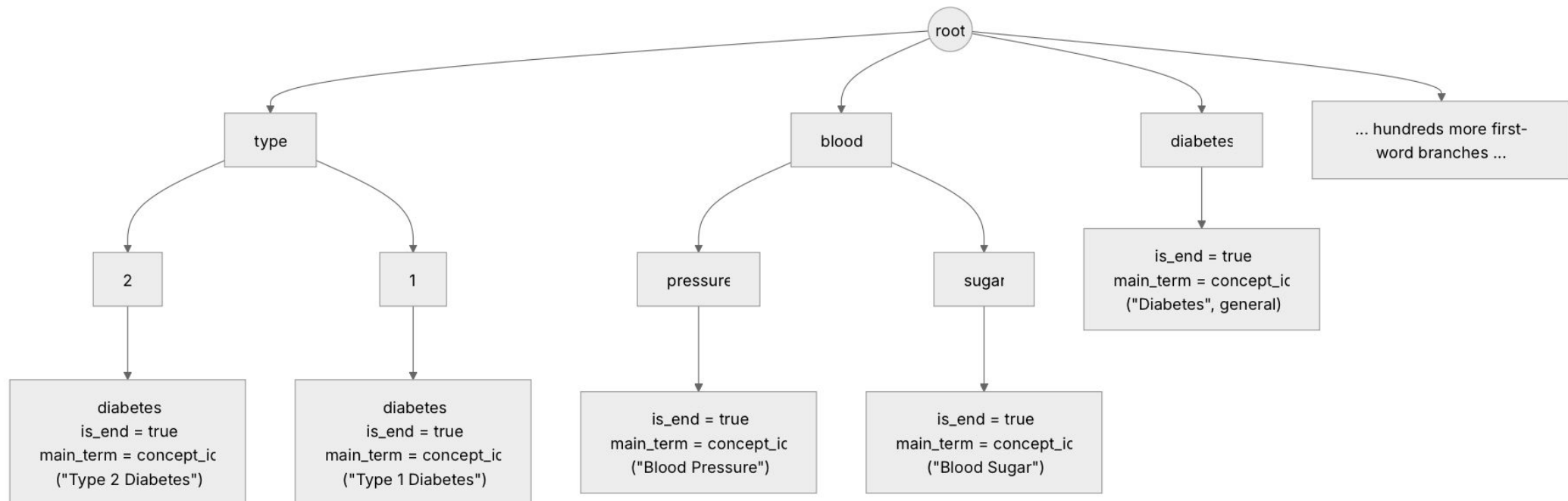
Why Fine-Tuning?

Instead of generating new medical information, the model learns to **select from trusted MedlinePlus content**, helping keep the explanation concise and grounded in the original source.

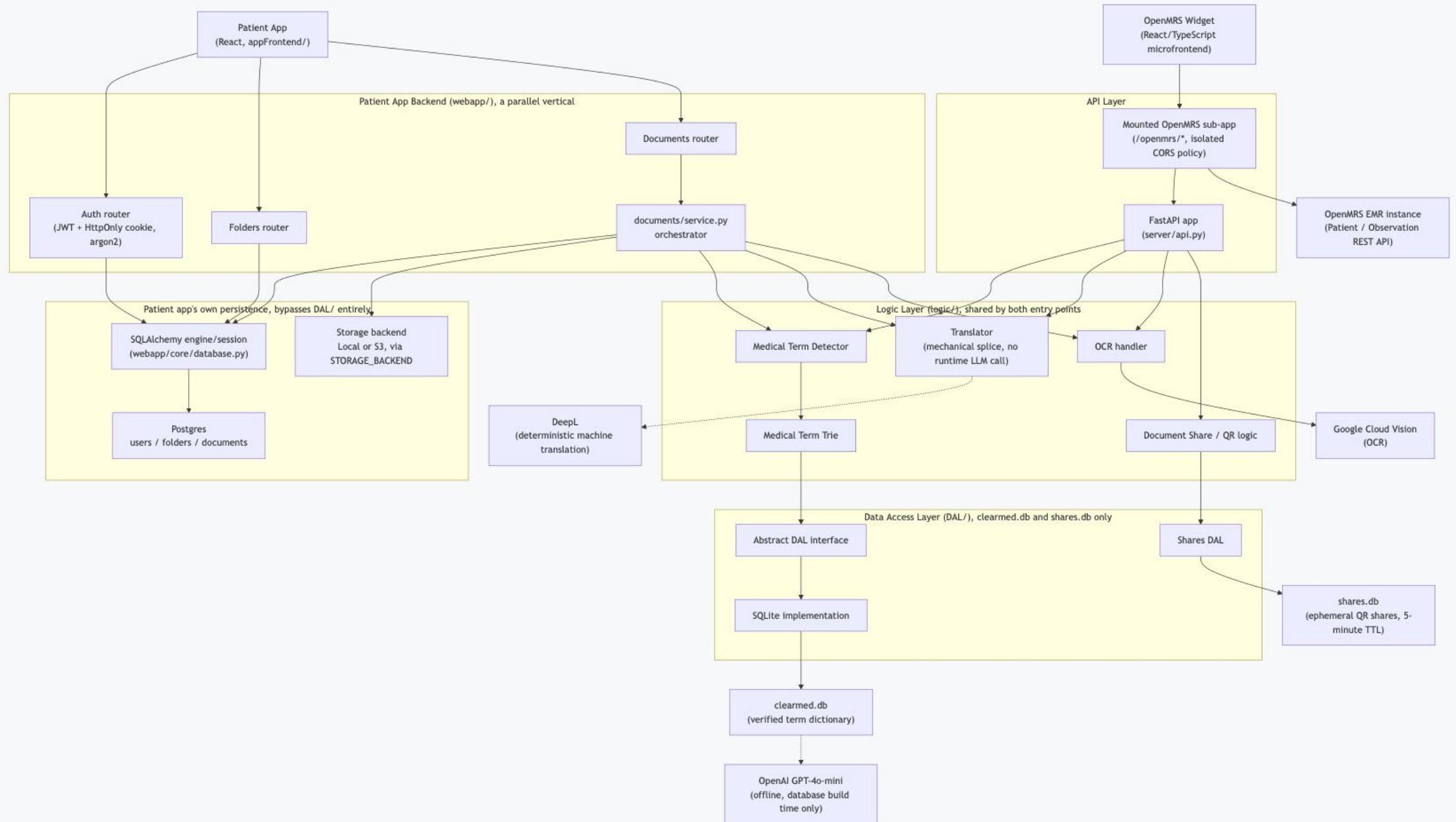
Prefix Trees to identify medical terms

Word-Level Indexing (Trie) – simplifying the term “blood pressure” and not “blood” - longest term lookup

The core search utilizes an in-memory word-level prefix tree, indexing over multi-word medical phrases. It relies on Python dictionary lookups at each node. it executes a longest-match, non-overlapping scan.



system Architecture-A decoupled 3-layer architecture



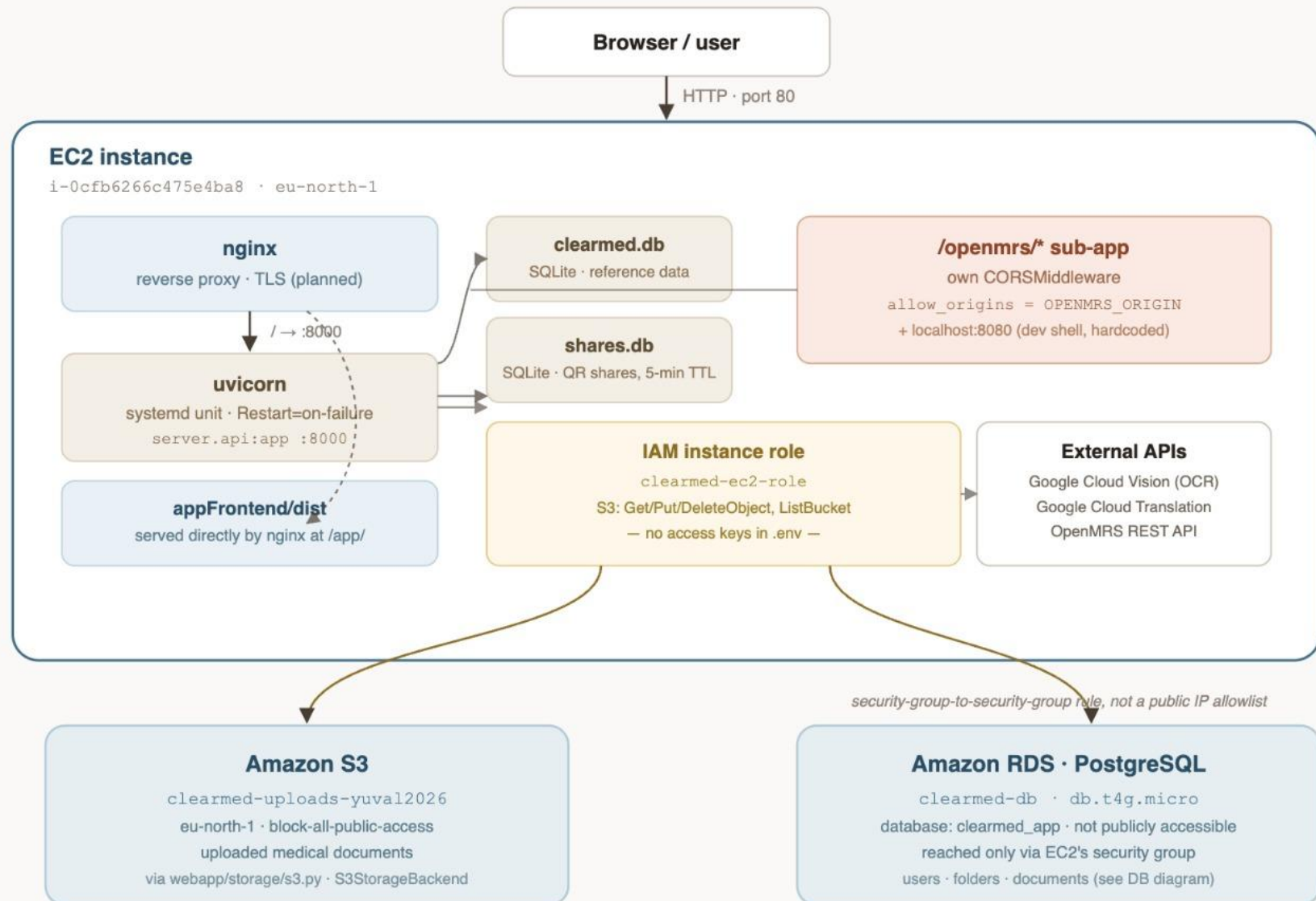
AWS Deployment Architecture

Single Region Infrastructure

Deployed on a single AWS EC2 instance ,
behind an Nginx reverse proxy routing to
Uvicorn.

IAM Roles: All AWS service access
(S3, Translation) is governed by
instance roles.

Network Isolation: The RDS
database is completely shielded from
public networks.



Database Architecture

Three Distinct Stores

Data is strictly segregated by its lifecycle and security requirements.

Stores are never merged or cross-queried.

PostgreSQL (RDS): Patient accounts, folders, and document metadata.

Migrated via Alembic.

clearmed.db (SQLite): Read-mostly medical reference dictionary. Rebuilt strictly offline.

shares.db (SQLite): Ephemeral QR document-share links with a strict 5-minute TTL.

